

Reverse Attribution: Identifying Counter-Evidence in Model Predictions

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Abstract—Reverse Attribution (RA) distinguishes negative feature attribution from a beneficial intervention in a frozen classifier. Given a known true class, it identifies negatively attributed feature groups and tests whether their reference replacement increases the true-class logit. An Attribution-Flip (A-Flip) score measures disagreement between grouped attributions for the leading classes. We evaluate BERT-base, RoBERTa-base, and ResNet-56 on 16-review IMDB and Yelp Polarity pilots and a 200-image CIFAR-10 subset. On CIFAR-10, A-Flip correlates weakly with confidence (Spearman $\rho = -0.156$); error-detection AUROC is 0.609 with a 95% bootstrap interval of [0.452, 0.766]. Both text pilots contain only correct predictions, precluding error-detection estimates. These findings support baseline-specific diagnostic analysis but do not establish calibrated uncertainty or comparative explanation superiority. Test-set checkpoint selection and finite-step attribution remain material limitations.

Index Terms—explainable machine learning, feature attribution, counter-evidence, Integrated Gradients, supervised model analysis

I. INTRODUCTION

Feature attribution describes input contributions to a model score relative to a reference. Attribution to a known true class can identify features associated with reduced support for that label. However, negative attribution does not guarantee that replacing a feature increases the true-class score: nonlinear interactions and reference dependence can separate attribution signs from intervention effects.

RA combines attribution scores with explicit interventions. It nominates negatively attributed groups, measures each independent reference replacement, and validates a candidate only when the true-class logit increases. This criterion concerns a fixed classifier’s response to an intervention, rather than causality in the data-generating process.

The Attribution-Flip (A-Flip) score is one-half the mean absolute difference between grouped attributions for the two leading classes. We examine its association with confidence and prediction error within each dataset.

We contribute a definition of counter-evidence validated by intervention, a grouped A-Flip statistic, and exploratory evaluation on text and images with individual attribution records. Counter-evidence analysis requires a known label and applies to correct and incorrect predictions. Unlabeled target selection is outside the study’s scope.

II. RELATED WORK

Integrated Gradients (IG) attributes a score difference along an input–reference path [1]. LIME fits interpretable local surrogates [2], and SHAP develops additive feature attribution [3]. These approaches already represent signed evidence. RA uses negative true-class scores to nominate interventions whose effects are then measured directly.

Counterfactual explanations seek changes associated with a desired prediction [4], with prototype-guided approaches encouraging interpretable alternatives [5]. RA’s reference replacements need not be plausible, label-preserving, or sufficient to change the prediction. Its *counterfactual pull* denotes the specified logit difference.

Sanity checks [6], text-explanation diagnostics [7], and controlled attribution benchmarks [8] distinguish visual plausibility from explanatory validity. A-Flip measures class-wise disagreement rather than perturbation robustness or probability calibration [9]. Matched explainer and human-subject comparisons are not evaluated here.

III. METHODOLOGY

A. Class Scores and Grouped Attribution

Let $f(x) \in \mathbb{R}^K$ denote the logits of a frozen classifier, y the known true class, and $\hat{y}$ and y_2 the largest- and second-largest-logit classes. Ties are resolved by ascending class index. Let E be the set of eligible feature groups and $d = |E| > 0$. A group can contain multiple scalar input coordinates.

For a reference b , coordinate j , and class k , straight-line IG is

$$a_j^{(k)} = (x_j - b_j) \int_0^1 \frac{\partial f_k(z)}{\partial z_j} \bigg|_{z=b+t(x-b)} dt. \quad (1)$$

For coordinate group G_i , define

$$\phi_i(x, f, k) = \sum_{j \in G_i} a_j^{(k)}. \quad (2)$$

Thus signed summation occurs before candidate filtering or absolute differences. The implementation uses Captum IG with Gauss–Legendre quadrature. The classifier must support differentiation with respect to its numerical input representation.

B. Feature Groups and Reference Inputs

For text, one group comprises the word-embedding coordinates of a tokenizer subword. Eligible positions exclude padding and special tokens. The reference sets eligible word embeddings to zero and preserves special-token embeddings. Attention masks, token-type IDs, and position IDs remain fixed during integration and interventions. Positional and subsequent embedding transformations remain active; zeroing a word embedding neither deletes a token nor zeros its contextual representation. For every analyzed example, an embedding-input adapter verifies that its unperturbed logits agree with the original token-ID forward pass.

For CIFAR-10, one group is a spatial pixel across all three RGB channels, giving $d = 32 \times 32 = 1024$. Inputs are standardized with channel means (0.4914, 0.4822, 0.4465) and standard deviations (0.2023, 0.1994, 0.2010). The reference is zero in standardized coordinates and therefore corresponds to dataset-mean RGB in display space. An intervention replaces all channels of one pixel. These groupings specify subwords and pixels rather than word spans or image regions.

C. Candidates and Intervention Validation

Negative true-class attribution defines the candidate set

$$C(x, y) = \{i \in E : \phi_i(x, f, y) < 0\}. \quad (3)$$

Let x_{-i} denote x with group i replaced by its reference. The counterfactual pull is

$$\Delta_i = f_y(x_{-i}) - f_y(x). \quad (4)$$

Pull is measured in logit units. Confidence, defined as the maximum softmax probability, is a separate predictive descriptor.

To limit evaluation cost, candidates are ordered by ascending signed attribution, breaking ties by feature index. The tested set T contains the first $q = \min(B, |C|)$ candidates for budget B . The validated set is

$$V(x, y) = \{i \in T : \Delta_i > 0\}. \quad (5)$$

The selected set S contains at most m members of V , ordered by descending pull and then ascending feature index. Our experiments use $B = 20$ and $m = 5$. Untested candidates have undefined pull and are excluded from validation. Zero and negative tested effects are retained in the records.

For $|S| > 0$, counter-evidence strength is

$$s(x, y) = \frac{1}{|S|} \sum_{i \in S} \Delta_i. \quad (6)$$

Strength is undefined for an empty selection. Dataset summaries average per-example strength over nonempty selections and report the denominator. Independently measured pulls do not imply additive effects under simultaneous replacement. Because positivity is imposed by selection, positive mean strength alone is not independent evidence of explanation quality.

TABLE I
DATASETS, OFFICIAL LABELED SPLITS, AND FROZEN CLASSIFIERS.
INTEGRATED GRADIENTS (IG) IS USED THROUGHOUT.

Dataset	Train	Test	Model	RA
IMDb	25,000	25,000	BERT-base	IG
Yelp	560,000	38,000	RoBERTa-base	IG
CIFAR-10	50,000	10,000	ResNet-56	IG

TABLE II
TRAINING SETTINGS RECORDED FOR THE EVALUATED CHECKPOINTS.
EPOCHS INDICATE SELECTED / PLANNED; LENGTH DENOTES TOKEN LIMIT
OR IMAGE DIMENSIONS.

Dataset	Batch	Epochs	Initial LR	Length
IMDb	16	2 / 3	$2 \cdot 10^{-5}$	512
Yelp	64	1 / 3	$2 \cdot 10^{-5}$	256
CIFAR-10	128	200 / 200	0.1	32×32

D. Attribution-Flip

We define A-Flip as

$$\alpha(x) = \frac{1}{2d} \sum_{i \in E} |\phi_i(x, f, \hat{y}) - \phi_i(x, f, y_2)|. \quad (7)$$

The denominator counts eligible groups rather than scalar coordinates. A-Flip is computed independently of whether C is empty. When the true class differs from both leading predictions, all three class-specific attribution vectors are computed.

A-Flip measures class-wise attributional disagreement. Its magnitude depends on logit scale, reference, grouping, input length, and model behavior. Consequently, its values are interpreted within a dataset and protocol; they do not provide a calibrated uncertainty scale or a basis for ranking modalities.

IV. EXPERIMENTAL DESIGN

A. Datasets and Classifiers

We consider binary sentiment classification on IMDb [10] and Yelp Polarity [11], and ten-class image classification on CIFAR-10 [12]. Table I lists the official labeled splits. IMDb's additional unlabeled partition is excluded.

The text classifiers use pooled BERT-base and RoBERTa-base representations [13], [14] followed by a 768-to-384-to-2 multilayer perceptron with ReLU and dropout. The image classifier is a CIFAR ResNet-56 [15]. Evaluation uses existing frozen checkpoints, with strict state-dictionary loading and matching architecture and tokenizer configurations.

B. Training Configuration and Model Selection

Table II summarizes checkpoint configurations. Text models used AdamW: weight decay 0.01, constant learning rate 2×10^{-5} , and all parameters trainable. No scheduler is implemented in the available trainer. Source defaults specify dropout 0.1 and gradient clipping at 1.0. Yelp records accumulation of one and an FP16 flag; execution of mixed precision is not independently established.

ResNet-56 used SGD (momentum 0.9, weight decay 10^{-4}) and MultiStepLR reductions of 0.1 at epochs 100 and 150. Training used random crops with padding four and horizontal flips; evaluation used no augmentation. Original training seeds and exact executed source revisions are unavailable.

Checkpoint selection repeatedly monitored the official test split. Table III therefore reports descriptive monitoring accuracy, not unbiased held-out estimates. The selected epochs are 2/3 (IMDb), 1/3 (Yelp), and 200/200 (CIFAR-10). Even the final-epoch CIFAR checkpoint was test-selected. Analysis of these fixed classifiers does not resolve this bias.

C. Sampling and Attribution Protocol

We sample 200 CIFAR-10 images and 16 reviews from each text dataset uniformly without replacement over the official test split. NumPy’s `default_rng` uses seed 42, and selected indices are sorted. Sampling does not condition on correctness or confidence. The text subsets are small exploratory pilots.

All examples use 50 IG quadrature steps, $B = 20$ candidate interventions, and $m = 5$ maximum selected features. Evaluation runs on CPU with two threads for text and four for vision; interpolation batches contain two and four points, respectively. Checkpoints remain fixed throughout attribution and intervention evaluation.

D. Statistical Analysis and Reproducibility

We report per-example A-Flip, candidate and validated counts, selected counts, and nonempty-selection strength. Associations use Spearman correlation with maximum softmax confidence and the top-two logit margin. Error-detection AUROC treats an incorrect prediction as the positive class and larger A-Flip as the score. Correct and incorrect A-Flip distributions are compared using a two-sided Mann–Whitney test.

We form percentile intervals from 1,000 bootstrap replicates, sampling examples with replacement (seed 42) for subsets of at least 20 examples. Undefined resamples are omitted; undefined statistics are unavailable. These intervals describe sample variation conditional on the frozen model, excluding training and model-selection uncertainty.

Per-example JSONL records retain sample identities, labels, logits, grouped attributions, all candidate and tested effects, selection counts, and IG completeness residuals. Manifests record checkpoint, dataset, tokenizer, and source hashes, settings, and environment versions. Tables and figures are derived from complete records with validated raw-result checksums. Qualitative cases follow a deterministic rule: the first misclassified example in sorted sample order, or the first example if all predictions are correct.

V. RESULTS

A. Counter-Evidence Statistics

Table IV summarizes the 232 analyzed examples. Mean candidate counts are 114.500 for IMDb, 42.750 for Yelp, and 504.755 for CIFAR-10. Among the budgeted candidates, the corresponding mean validated counts are 10.250, 8.500, and 9.520. Mean selected counts are 5.000, 4.688, and 4.785;

TABLE III
STORED CHECKPOINT MONITORING ACCURACY (FRACTION).
OFFICIAL-TEST CHECKPOINT SELECTION MAKES THESE DESCRIPTIVE
VALUES RATHER THAN UNBIASED HELD-OUT ESTIMATES.

Dataset	Checkpoint epoch	Stored accuracy
IMDb	2	0.935840
Yelp	1	0.976237
CIFAR-10	200	0.932800

TABLE IV
MEAN RA STATISTICS FOR FROZEN CLASSIFIERS. SELECTED COUNT IS AT
MOST FIVE PER EXAMPLE; STRENGTH AVERAGES NONEMPTY SELECTIONS
IN LOGIT UNITS. TEXT SUBSETS ARE EXPLORATORY PILOTS.

Dataset	n	A-Flip	Selected	Strength
IMDb	16	0.045	5.000	0.068
Yelp	16	0.083	4.688	0.204
CIFAR-10	200	0.571	4.785	1.022

strength is defined for every example. These differences demonstrate that candidate, validated, and selected sets have distinct empirical sizes.

Mean A-Flip is 0.045 for IMDb, 0.083 for Yelp, and 0.571 for CIFAR-10, with population standard deviations of 0.015, 0.051, and 0.269. These values describe separate model/reference configurations and are not interpreted as a cross-domain ordering of uncertainty.

B. Confidence and Prediction Error

The CIFAR-10 subset contains 10 errors. A-Flip correlates weakly and negatively with confidence ($\rho = -0.156$, 95% CI $[-0.281, -0.019]$) and logit margin ($\rho = -0.165$, 95% CI $[-0.287, -0.028]$); see Table V and Fig. 1. Mean A-Flip is 0.568 for correct predictions and 0.628 for errors. The two-sided Mann–Whitney comparison gives $p = 0.247$.

Error-detection AUROC is 0.609, with a 95% bootstrap interval of $[0.452, 0.766]$. The wide interval includes chance performance and does not establish reliable error detection. Both text pilots contain zero errors, making their error AUROC and incorrect-group comparisons undefined. Confidence correlations are 0.006 for IMDb and 0.135 for Yelp; these observations do not support a consistent negative association across datasets.

C. Qualitative Interventions

Figure 2 shows CIFAR-10 test example 226, the first misclassification in the sampled order. The attribution map is the signed true-class IG summed over channels. The modified image marks selected intervention locations by replacing those pixels with dataset-mean RGB. Individual pulls were measured separately; the displayed simultaneous replacement does not report a joint effect.

Figures 3 and 4 show the first examples of the all-correct text pilots, IMDb index 2147 and Yelp index 3265. Bars report the independent true-class logit changes for selected subword positions. Token strings and tokenizer-specific prefix

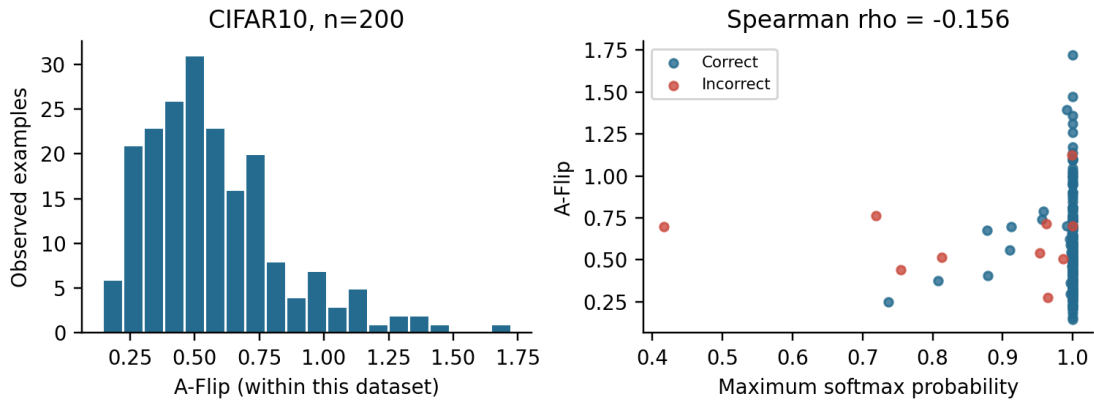


Fig. 1. A-Flip on the 200-image CIFAR-10 subset. Left: empirical distribution. Right: association with maximum softmax probability, with one point per example and colors indicating prediction correctness.

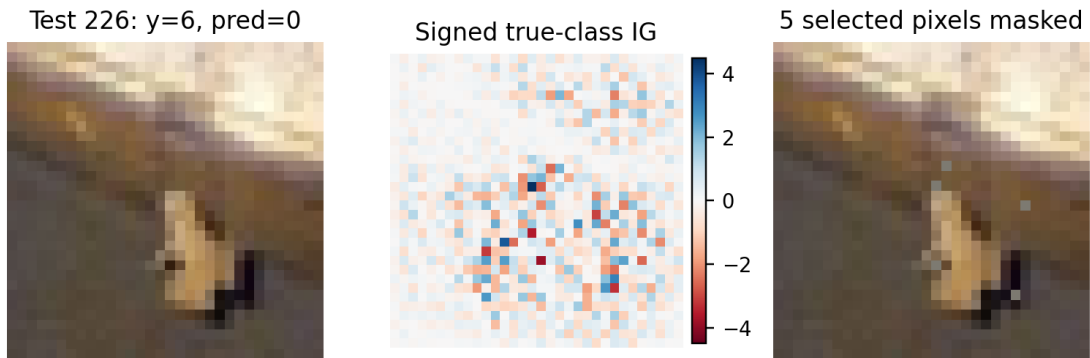


Fig. 2. CIFAR-10 test example 226. Left: original input and labels. Center: signed true-class IG aggregated over RGB channels. Right: selected pixel locations replaced with dataset-mean RGB. Pulls are evaluated independently for each selected pixel.

TABLE V
EXPLORATORY WITHIN-DATASET ASSOCIATIONS. AUROC USES LARGER A-FLIP TO PREDICT ERROR; UNDEFINED VALUES ARE SHOWN AS $-$.

Dataset	Errors	ρ_{conf}	ρ_{margin}	AUROC
IMDb	0	0.006	0.006	$-$
Yelp	0	0.135	0.126	$-$
CIFAR-10	10	-0.156	-0.165	0.609

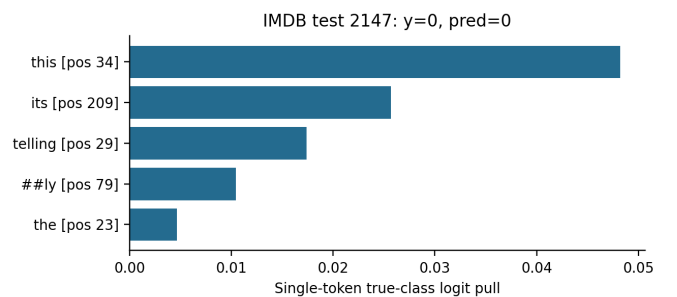


Fig. 3. IMDb test example 2147: independent zero-word-embedding interventions at selected subword positions, measured in true-class logit units.

markers identify model input units; they are not semantic annotations. The examples illustrate baseline-specific counter-evidence within correct predictions.

D. Numerical Reliability

Absolute IG completeness residuals have median and maximum values of 0.327/3.397 logits for IMDb, 0.108/0.376 for Yelp, and 0.285/2.323 for CIFAR-10. Increasing quadrature from 50 to 200 steps on the first five sampled CIFAR images changed A-Flip by 0.019–0.226% and reduced each case’s maximum residual. This diagnostic does not establish full-sample convergence; all reported aggregates use 50 steps.

VI. DISCUSSION AND LIMITATIONS

Interpretation. Validation establishes a positive effect under a specified intervention, not explanatory superiority, semantic causality, or label preservation. Matched explainer comparisons and controlled semantic evaluation remain necessary.

Reference and representation. Zero word embeddings and dataset-mean pixels may produce off-manifold inputs. Subwords and pixels can be less interpretable than spans

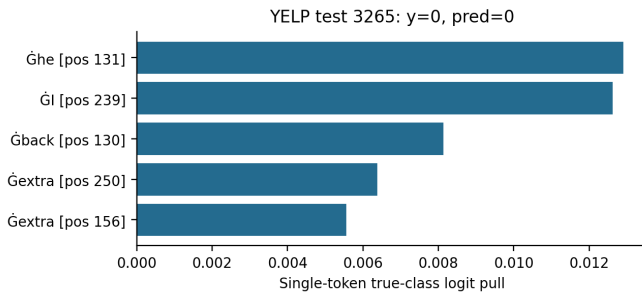


Fig. 4. Yelp Polarity test example 3265: independent logit effects at selected subword positions under the same intervention protocol.

and regions. Reference and grouping dependence limit A-Flip comparisons across configurations.

Selection and sampling. Official-test monitoring biases the frozen checkpoints. Few CIFAR errors limit precision, and the text pilots cannot support error-detection estimates. Bootstrap intervals exclude variation from training seeds, checkpoint selection, and references.

Numerics. Finite-step IG yields nonzero completeness residuals. Quadrature and floating-point effects can change near-zero signs; Eqs. (3) and (5) apply strict inequalities without a numerical threshold. Implementation checks do not establish convergence of every explanation.

VII. CONCLUSION

Reverse Attribution distinguishes negative-attribution candidates from features whose reference replacement increases the true-class logit. A-Flip quantifies grouped attribution disagreement between leading classes. Frozen-classifier experiments demonstrate the procedure and a weak CIFAR-10 confidence association, with substantial uncertainty in error detection. Broader utility requires larger independent evaluations, reference-robustness studies, matched explainer comparisons, and clean model selection.

DATA AND CODE AVAILABILITY

The accompanying implementation includes evaluation settings, per-example records, artifact manifests, and table/figure generation scripts. Execution requires the exact checkpoint and dataset artifacts identified by their hashes. These local artifacts are not automatically distributed with the repository; access must be obtained from the author.

APPENDIX A

ATTRIBUTION AND INTERVENTION CAN DISAGREE

Consider the interacting score function

$$f_0(x) = -3x_1 + 4x_1x_2 - 3x_2 - 2x_3. \quad (8)$$

At $x = (1, 1, 1)$ with a zero reference, IG is $(-1, -1, -2)$. Replacing either of the first two coordinates gives pull -1 ;

replacing the third gives $+2$. All coordinates are candidates, but only the third is validated. This example is also exercised by the implementation tests.

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